AI-Based Campus Security System

Final Year Project

PROPOSAL

Supervisor: Nadeem Sarwar

Submitted By (BSCS-F25-010)

M. Mansoor Fareed (03-134222-069)

Qazi Taha Majid (03-134222-090)

Umer Javaid (03-134222-110)

Bachelor of Science in Computer Science

Department of Computer Sciences

Bahria University Lahore Campus

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Version	Date	Remarks
1.0		First Complete Submission to FYP Office

Abstract

Ensuring campus security is a growing challenge in universities. Manual identity verification methods, such as guards checking ID cards and fingerprint scanners for attendance, are prone to errors, delays, and security breaches. Intruders may enter using fake or borrowed cards, and monitoring the entry and exit of students, faculty, and visitors becomes inefficient. Parents and university authorities also lack a real-time mechanism to track student presence and safety within the campus. Therefore, a robust AI-driven system is required. This project proposes an AI-powered gate security prototype integrating facial recognition, iris scanning (for veiled/masked students), university card scanning and CNIC verification for visitors. The system cross-checks faces and ID data against the university database in real time. It logs check-in/check-out times, notifies parents and students via a portal and SMS, and generates attendance reports for HR and faculty. A multi-role web portal will provide dashboards for students, faculty, parents, directors, and security staff. Hardware includes face recognition terminals, iris scanners, university card scanners, CNIC document scanners, and edge AI devices (Intel NUC). Software will use Python (FastAPI, React frontend), PostgreSQL, and InsightFace for recognition.

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1. Introduction

Campus safety has become a fundamental concern for higher education institutions, particularly in large campuses where thousands of individuals (students, faculty members, staff, and visitors) enter and exit on a daily basis. Traditional methods of ensuring security, such as relying on guards, manual identity card inspections, or fingerprint-based attendance systems, are increasingly proving to be insufficient in meeting modern security challenges. These approaches not only slow down the entry and exit process due to their time-consuming nature but are also highly susceptible to human error, intentional misuse, or even security breaches. For instance, manual inspections are limited by the attentiveness and accuracy of the guards, while fingerprint systems can face hygiene concerns, duplication risks, and technical failures. Consequently, universities face growing risks related to unauthorized access, fake ID cards, impersonation, and inefficiency in record keeping. These limitations highlight the urgent need for a more intelligent, automated, and error-resistant system that ensures both security and convenience for campus stakeholders.

The proposed AI-Based Campus Security System [10] addresses these challenges by introducing a fully automated prototype to be installed at the university gate. Unlike conventional systems, this solution integrates multiple biometric and verification modalities into a single, unified security framework. The system employs advanced **facial recognition algorithms [1]** for primary verification, **iris scanning [2]** for cases where individuals may wear masks or veils, **OCR-based university ID card scanning [3]** for institutional validation, and **CNIC verification** for visitors, which can be cross-checked against criminal record databases for added security. Beyond identity verification, the system introduces **real-time attendance logging** directly linked to the university database, enabling precise tracking of entry and exit times. It further enhances accountability by sending **instant notifications [4] to parents** whenever a student enters or leaves the campus. Additionally, a **multi-role digital portal [5][8]** is included, providing tailored access for students, faculty, parents, HR, security staff, and administrators. This ensures transparency, effective monitoring, and better decision-making. By reducing reliance on manual checks and embedding artificial intelligence into campus security protocols, the project aligns with the broader vision of building smart campuses institutions that are secure, technologically advanced, and capable of handling the growing scale of academic communities.

2. Problem Description

2.1. Primary Scope

The system aims to secure campus gates by automating identity verification and logging processes. It answers the following:

- **What?** Automate entry/exit validation of students, staff, faculty, and visitors using AI-based facial and iris recognition combined with ID scanning.
- **Why?** To reduce errors, prevent intruders, ensure parental/student notifications, and improve security efficiency.
- **How?** By integrating facial recognition, smart card scanning, CNIC scanning, and real-time database validation with a web-based portal.

2.2. Final Deliverable and Beneficiaries

The deliverable will be an integrated hardware-software system including web portal.

2.2.1. Hardware Deliverables:

IP Cameras (2 units): One for facial recognition, one for ID card OCR (1080p, IR, 30fps).
Edge Computer: Local AI processing for real-time decisions.
Edge Router: Secure communication between devices and database.
Retina Scanner (USB): Secondary biometric check for masked/veiled individuals.
Hardware Frame: Mounting structure for cameras and scanner.
Miscellaneous: Power units, connectors, and cables.

2.2.2. Software Deliverables:

Facial Recognition: InsightFace (ArcFace) with anti-spoofing.
ID Card Scanning: Tesseract/Google Vision OCR for verification.
CNIC Scanning: OCR with criminal record check integration.
Retina Scanning Software: Iris ID iCAM SDK.
Database: PostgreSQL for user records and logs.
Backend API: FastAPI with JWT authentication and role-based access.
Notification System: SMS/Email alerts for parents and admins.
User Portals: Students, faculty, parents, HR, security, and admin.
Event & Paid Pass System: Manage and verify digital event passes.

Direct beneficiaries: Students, faculty, staff, and parents (attendance & safety tracking).
Indirect beneficiaries: University administration, HR, security staff, and event organizers.

2.3. Optional Scope

Optional scope includes event payment integration [6] and criminal record verification through CNIC database cross-checking.

2.4. Objectives

- To develop a gate-based security prototype using facial recognition and RFID card scanning.
- To implement iris scanning for veiled/masked students.
- To enable real-time face-to-card and face-to-database cross-checks.
- To create a multi-role web portal for students, parents, faculty, HR, and security.
- To automate attendance marking and notify parents of entry/exit times.
- To integrate visitor CNIC scanning and alert system for potential security risks.

2.5. Novelty

Previous campus security systems primarily relied on fingerprint attendance or manual card checking. This system is novel because it integrates multi-modal biometrics (face + iris + card + CNIC), real-time parental notifications, and a multi-role portal with event and payment features. There is no existing FYP at Bahria University that combines all these features into a unified solution.

3. Methodology

3.1. Facial Recognition:

- Use **OpenCV + Deep Learning-based models (InsightFace / ArcFace embeddings)** for robust facial feature extraction and matching.
- Anti-spoofing module (liveness detection via blink/skin texture analysis) implemented directly on the IP camera feed processed on the **Intel NUC**.

3.2. Retina Scanning:

- Integrate the **USB Retina Scanner (with SDK)** to authenticate veiled or masked students where facial recognition may fail.
- The retina scan will generate a unique biometric key stored securely in the database and matched at the gate.

3.3. Card Reading (Simulated/Basic Implementation):

- Since no dedicated RFID/NFC reader is included in the prototype budget, card scanning will be simulated using a **QR/barcode printed on university cards**, scanned through the **IP camera** and decoded using **OpenCV + ZBar library**.
- This keeps the solution low-cost and hardware-light.

3.4. CNIC Scanning (Visitors):

- CNIC details will be captured using the **IP camera** (high-resolution image).
- OCR will be applied using **Tesseract OCR** or **Google Vision API** to extract CNIC number for visitor verification.

3.5. Database:

- **PostgreSQL** will store facial embeddings, retina scan data, card/visitor data, and access logs [7].

3.6. Backend:

- **FastAPI (Python)** for gate-side API and role-based authentication.
- **JWT (JSON Web Tokens)** for secure multi-role login (student, faculty, parent, security, admin).

3.7. Frontend:

- **React.js** portal with dashboards for students, faculty, HR, parents, and security staff.

- Features include entry/exit logs, event management, fee pass payments, and real-time alerts.

3.8. Notifications:

- **Firebase Cloud Messaging (FCM)** for real-time push notifications.
- **Twilio SMS API** for parent alerts on student entry/exit.

3.9. Hardware Integration:

- **IP Cameras (1080p, IR/night vision)** for capturing faces, QR/barcodes, and CNIC images.
- **Intel NUC** for edge processing [9] of AI tasks (facial recognition, OCR, database sync).
- **USB Retina Scanner + SDK** for secure biometric fallback in case facial recognition fails.

3.10. Flowchart:

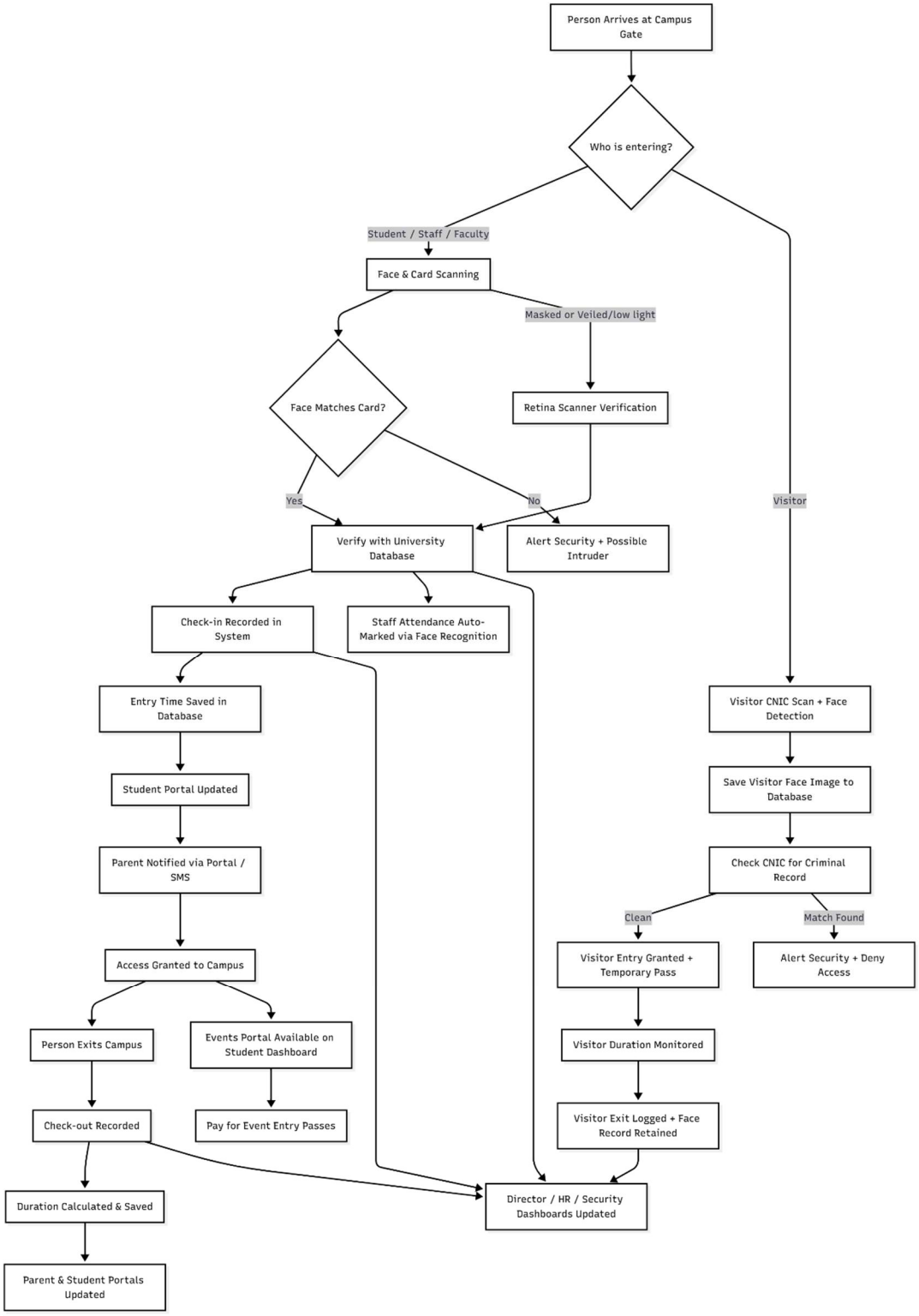

Figure 3-1: Functional Flowchart

4. Feasibility Plan

4.1. Resource Requirements:

Hardware: Face recognition terminal, iris scanner, card reader, CNIC scanner, NAS storage.
Software: Python, FastAPI, React, PostgreSQL, InsightFace, Tesseract, Firebase/Twilio.

4.2. Expertise of the Team:

Team members have completed OOP, DSA, AI, and Computer Vision courses. They have hands-on experience in Python, ML frameworks, and web development.

4.3. Tools / Technology:

Languages: Python, React, PostgreSQL

AI Tools: PyTorch, InsightFace

APIs: Twilio, Firebase, Google Vision OCR

Cloud/Edge: Intel NUC hardware for inference

4.4. Budget(Prototype):

Items	**Cost (PKR)**	**Remarks**
Two IP Cameras	20,000	1080p, IR Night Vision
Edge Computer & Internal USB	20,000	For local computation
Retina Scanner	15,000	USB-enabled
Hardware Structure Frame	5,000	Mounting cameras
APIs	3,000	Google vision & Twilio sms
Miscellaneous	5,000	Cables, connectors

4.5. Risks Involved:

Hardware availability: Specialized scanners may be costly.

Technical risks: Performance issues with masks/poor lighting; mitigated using iris scanning and IR cameras.

5. Key Milestones and Schedule

5.1.1. Key Milestones

S. No.	S. No. of Predecessor or Milestone	Key Milestone Name / Description	Duration (person-hours)
1	1	Initial research and requirement analysis	60
2	2	Proposal Defence	30
3	3	Requirement gathering	30
4	4	Environment setup + Facial recognition module	135
5	5	University card scanning + Face-card comparison	150
6	6	CNIC scanning (OCR) + Retina scanner integration	90
7	7	Database integration + Notification system + Portal development	135
8	8	Test and refine system	114
9	9	Final testing and project submission	120

5.2. Gantt Chart

Following is the timeline divided into milestones mentioned

Activity	Weeks																	
	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18
Milestone 1									Midterm Exam Week									Exam Week
Milestone 2																		
Milestone 3																		
Milestone 4																		
Milestone 5																		

Figure 5-1: SEMESTER 7(Fall 2025)

Activity	Weeks																	
	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18
Milestone 6									Mid Exam Week									Exam Week
Milestone 7																		
Milestone 8																		
Milestone 9																		

Figure 5-2: SEMESTER 8(Spring 2026)

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